

Statement of Research

Interfacial Processes for Critical Resource Recovery at the Food–Energy–Water Nexus

Clean water and secure material supplies are often treated as separate challenges, yet they converge in water, waste streams, and contaminated soil. These streams contain valuable critical materials, but the resources are dilute and mixed with salts, metals, organic matter, and persistent contaminants. Conventional treatment often consumes energy and chemicals across the entire bulk stream while recovering little value. My research addresses this selectivity challenge by studying the biochemical interfaces where catalysts, electric fields, and microorganisms govern transport and reaction. Together, these processes support advanced water reuse, domestic critical-material production, and soil remediation.

Previous Research

My previous research has advanced the biochemical interfaces at the **food–energy–water nexus** through **resource recovery, remediation, and mitigation**. I have participated in projects funded by the U.S. Department of Energy’s Advanced Materials and Manufacturing Technologies Office, NASA LaSPACE, the Gates Foundation’s Water, Sanitation, and Hygiene program, the Heintzelman Carbon Sequestration Fund, and Southwest Airlines. At Princeton, I advanced carbon redirection by engineering temperature-controlled micro-redox environments that suppressed methanogenesis and redirected wastewater fermentation toward volatile fatty acids. This DOE-supported project also gave me experience leading a four-researcher team and managing sponsor deliverables. At Yale, I extended carbon redirection from wastewater to biomass by developing a one-pot electrochemical route that converted lignin-derived catechol to the nylon precursor dimethyl adipate. I also secured a Southwest Airlines funding as the principal investigator for biomass-to-aviation-fuel research. At Rice, I focus on critical-resource capture and soil remediation by leading Gates Foundation-supported scale-up of autonomous system for reliable decentralized water reuse and investigating contaminant transport.

Future Research Vision

My independent program will continue to develop **interfacial biochemical processes** that recover critical materials and renewable chemicals from water and waste streams. Rather than treat the entire bulk solution, I will control electric fields, ligand chemistry, gas–water interfaces, and microbial electron flow at the sites where separation and reaction occur. My program has three connected thrusts: (1) Electrified micro-interface **resource recovery** from diverse domestic resources; (2) nanobubble-enabled mobilization and **remediation** of PFAS from soil; and (3) **mitigation** of electrochemical disinfection by fouling control.

These projects will advance the technologies at the food-energy-water nexus, secure funding from multiple agencies (listed in Fig. 1), and open collaboration opportunities across multiple departments.

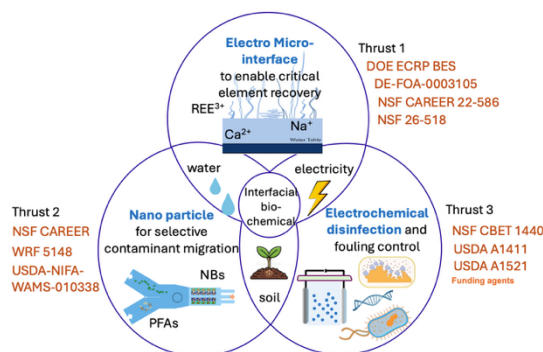


Fig 1. Overview of future research plan and their funding opportunities (listed in orange notes)

Thrust 1. Electrified Micro-Interfaces for Critical-Mineral Recovery

Critical minerals, including lithium and rare-earth elements, are essential to electric vehicles, wind turbines, electronics, and other advanced technologies, yet the United States remains dependent on concentrated supply chains. Waste electronics, mine drainage, geothermal brines, and oil- and gas-produced water provide promising domestic sources. However, target minerals occur at low concentrations alongside abundant salts and competing metals. Recent advances have been made by electrodialysis technologies with the integration of monovalent ion selective membrane. However, a central challenge is achieving selectivity for target critical minerals in complex, high-salinity water streams.

I will develop **electrified micro-interfaces** that integrate **selective recovery** of critical-mineral ions, **capillary-driven** water transport, and **Joule-heating-enabled** evaporation on a single conductive faber surface

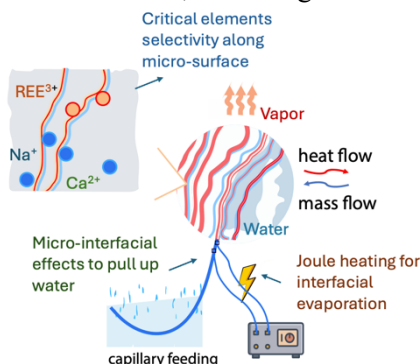


Fig 2. Selective capture, field-driven transport, and localized mineral recovery

(Fig. 2). This work builds on my prior experience fabricating electroactive materials for capillary transport and localized Joule heating. To selectively recover target critical minerals, I will **functionalize conductive fiber scaffolds** with phosphate-containing ligands, metal-organic-framework coatings, or ion-imprinted polymers designed to **preferentially bind specific metal ions**. This approach builds on my previous expertise in fabricating MOF-based materials. Because ion capture and concentration occur at accessible fiber interfaces rather than within a dense membrane, this architecture may reduce internal blockage and fouling in complex streams.

Thrust 2. Nanobubble-Enabled PFAS Mobilization for Soil Remediation

PFAS contamination in soil and groundwater is difficult to remediate because the interfaces that govern PFAS retention and release cannot be observed directly with conventional column measurements. PFAS molecules partition among water, mineral surfaces, organic matter, and air-water interfaces. Existing pump-and-treat systems remove PFAS from water but leave a large amount of water with low concentration of PFAS. A treatment method that can mobilize and concentrate PFAS would reduce the volume requiring final destruction.

I hypothesize that **nanobubbles can serve as mobile carrier for PFAS removal and soil remediation**. Their high interfacial area, persistence, and surface charge may allow them to collect surface-active PFAS and transport it through connected soil pores. When applying electrical field, nanobubble can carry target contaminant on the air-gas surface and transport toward an electrically controlled direction. I will use **soil-on-a-chip devices** and **super-resolution microscopy** to observe PFAS-nanobubble interactions based on my previous expertise in microfluidic-device design (Fig. 3). These experiments will reveal how water chemistry, PFAS structure, mineral composition, pore geometry, and electric fields influence bubble movement and PFAS attachment.

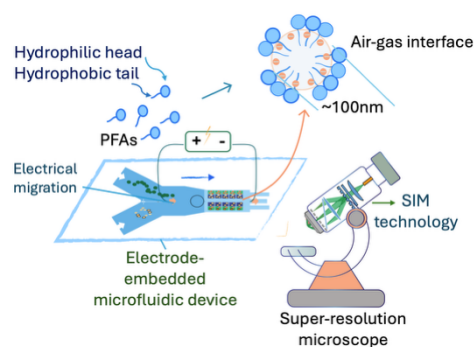


Fig 3. Observe and control mobile nanobubble interfaces for PFAS remediation

Thrust 3. Reliable Fouling-Control in Electrochemical Disinfection

Reliable water disinfection is fundamental to preventing waterborne disease, enabling safe water reuse, and protecting public health, particularly in communities without dependable centralized treatment. Electrochemical disinfection offers an attractive alternative because it uses electricity to generate reactive disinfecting species in situ or directly inactivate microorganisms at electrode interfaces. However, fouling by organic matter, salts, and microorganisms can block active surfaces, alter current distribution, increase energy demand, and reduce disinfection reliability.

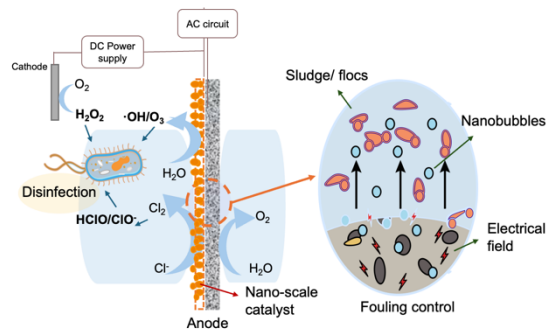


Fig 4. Electrochemical disinfection by controlling fouling at electrode-water interfaces

I will investigate **alternating electric fields as a strategy for controlling fouling in electrochemical disinfection (Fig. 4)**. Time-lapse optical and fluorescence imaging will reveal the formation, growth, and removal of organic films, mineral deposits, biofilms, and gas bubbles. By integrating imaging with electrochemical measurements, I will link local fouling dynamics to current distribution, disinfectant generation, pathogen inactivation, and energy consumption. I will also develop **electrochemical lab-on-a-chip devices to observe the fouling and disinfection processes in situ**.

Together, these three thrusts integrate electrified micro-interfaces for critical-material recovery, lab-on-a-chip imaging and nanobubble transport for persistent-contaminant remediation, and strategies to mitigate fouling-control in electrochemical disinfection. I will leverage my expertise in environmental microbiology, electrochemistry, interfacial transportation, and reactor design to advance water security, soil restoration, domestic material supplies, and low-carbon manufacturing.

Selected References

[1] Lu et al., Green Chem. 2026, 28, 12428-12437. [2] Lu et al., Water Res. 2024, 265, 122306. [3] Lu et al., Resour. Conserv. Recycl. 2024, 203, 107407. [4] Lu et al., Biosens. Bioelectron. 2020, 167, 112491. [5] Lu et al., Chem. Eng. J. 2019, 373, 1072-1080.